

Redesigning Global Operations Intelligence

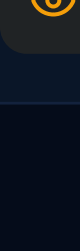
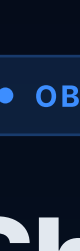
Transforming a legacy manufacturing dashboard into a predictive decision platform for executive operations management at Velocity Motors.

ROLE	TOOL	CONSTRAINT	FOCUS AREA
UI/UX Designer	Figma	Power BI Feasibility	Predictive Intelligence

PROBLEM STATEMENT

The Dashboard Was Reporting — Not Enabling Decisions

Velocity Motors relied on a legacy Power BI dashboard to monitor plant performance. But the system supported **reporting, not strategic action**.

 Downtime prediction accuracy Below industry standard	75%
 Fragmented data silos Disconnected systems	6+
 Financial impact visibility No revenue linkage	None
 Proactive risk alerts Purely reactive system	0
 Executive clarity score Too granular for leadership	Low

We have data everywhere, but we can't see what's actually happening. By the time we know there's a problem, it's already costing us millions.

VP of Global Operations
Velocity Motors — Assumption-based persona

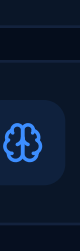
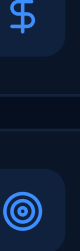
LEGACY DASHBOARD — BEFORE

OEE Rate	Throughput	Defect %
Uptime	MTTR	Inventory
No predictive intelligence. No financial context. No action pathway.		

OBJECTIVE

Shift From Reactive Monitoring to Predictive Intelligence

Design a modern executive dashboard that transforms passive reporting into proactive, actionable intelligence, at the right level of detail for leadership decision-making.

 Improves visibility across global plants	✓
 Enables early risk detection through predictive intelligence	✓
 Connects operational metrics to financial impact	✓
 Supports prioritization and executive action	✓
 Remains fully feasible within Power BI ecosystem	✓

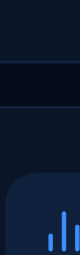
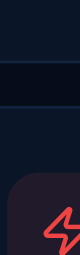
RESEARCH

Understanding Manufacturing Decision-Making

Without direct stakeholder access, I conducted secondary research across manufacturing intelligence and executive UX literature to ground the design in real operational workflows.

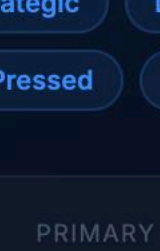
RESEARCH AREAS

- > Manufacturing KPIs (OEE, throughput, defect rate)
- > Predictive maintenance frameworks
- > Enterprise BI dashboard patterns
- > Executive operational workflows

 Executives scan first, analyze second Information hierarchy must support 5-second situational awareness before deep analysis.	 Financial impact drives prioritization Leaders act faster when metrics are anchored to revenue risk, not abstract KPIs.	 Alerts must be clear, not overwhelming Alert fatigue causes executives to ignore warnings. Only critical, contextual alerts work.
 Comparison across plants is critical Regional benchmarking enables relative prioritization and resource allocation decisions.	 Prediction without action is ineffective Risk scores must be paired with specific recommended actions to drive outcomes.	

USER PERSONA

VP of Global Operations

David Chen VP, Global Operations Strategic Data-Driven Time-Pressured Global Mindset PRIMARY GOAL Fast, actionable, financially contextualized intelligence	RESPONSIBILITIES - Oversee 12 global assembly plants - Monitor production performance - Reduce downtime cost - Prioritize maintenance investment ASSUMPTION-BASED PERSONA Developed through secondary research on manufacturing decision-making workflows and enterprise BI usage patterns. With stakeholder access, this would be validated through interviews and contextual inquiry.	PAIN POINTS - Too many disconnected reports - Delayed risk identification - Lack of financial clarity - Hard to compare regional performance
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USER FLOW

Executive Decision Flow

This flow guided the entire layout hierarchy — from top-level overview to specific action.

01 Open Dashboard Executive arrives with intent to get situational overview	02 Scan Global KPIs 5-second health check of key operational metrics	03 Identify Risk Alert Critical alert surfaces automatically with context	04 Compare Affected Plant Regional benchmarking positions the plant in context	05 Analyze Predictive Risk Deep-dive into failure probability and contributing signals	06 Review Financial Impact Downtime cost estimate anchored decision in revenue terms	07 Take Preventive Action AI recommendation with one-click escalation pathway
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INFORMATION ARCHITECTURE

Four-Layer Structure

Designed top-to-bottom for progressive decision-making, from 5-second scan to specific preventive action.

01 Executive Overview Immediate operational health check, Global OEE, downtime cost, active risk alerts, production vs target.	02 Regional Comparison Identify where performance gaps exist, OEE comparison by plant, throughput trends by shift, interactive filters.	03 Predictive Maintenance Understand why failure might occur, Vibration vs risk visualization, downtime probability score, financial impact.	04 Prioritization & Action Rank plants and recommend next steps, Performance ranking, risk indicators, AI-powered preventive action.
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CORE INNOVATION

Connecting Operations to Business Impact

Instead of showing isolated metrics, the design connects raw operational data all the way through to executive action in a single, coherent flow.

1 Sensor Data 847 IoT sensors	2 Failure Probability ML anomaly detection	3 Downtime Cost Financial modeling	4 Recommend Action AI-generated playbook
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This chain transforms **passive reporting** into **proactive intelligence**, giving executives the ability to act before failure occurs, with full awareness of financial consequences.

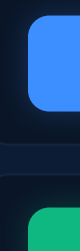
DESIGN PRINCIPLES

Five Guiding Principles

01 Decision-First Architecture Every layout choice is anchored to the executive decision flow, not technical data hierarchy.	02 Progressive Disclosure Overview → Analysis → Prediction → Action. Users drill deeper only when they need to.	03 Financial Contextualization Operational signals are always connected to revenue impact, making priorities self-evident.
04 Cognitive Clarity Clear visual hierarchy, high contrast, and minimal decorative elements reduce cognitive load.	05 Enterprise Feasibility All design decisions are validated against Power BI's capabilities, fully implementable as designed.	

VISUAL SYSTEM

Color Logic & Typography

COLOR SYSTEM		
 Deep Navy #060D1F	Structure	
 Electric Blue #3D8EFF	Actions	
 Success Green #10B981	Healthy	
 Warning Amber #F59E0B	Warning	
 Critical Red #EF4444	Critical	
 Slate Text #94A3B8	Secondary	

TYPOGRAPHY	
PRIMARY — INTER	
Global OEE 84.7%	
Aa Bb Cc Dd 0-9 @!#	
INTER 800: HEADLINES	INTER 700: KPI VALUES
Operations	\$2.4M
INTER 800: LABELS	INTER 400: BODY TEXT
CRITICAL	Risk analysis
Design Rationale Dark theme supports high data density and minimizes eye strain in operational control room environments. Color coding follows universal traffic-light semantics for instant comprehension.	

EXPECTED IMPACT

From Monitoring Performance to Optimizing It

Projected outcomes based on benchmark data from enterprise BI transformation case studies in the manufacturing sector.

40% Faster decision-making Reduction in time from alert to executive action	92% Prediction accuracy target Up from 75% with legacy reporting system	35% Risk exposure reduction Expected reduction in unplanned downtime costs	12 Plants unified Single platform for all global assembly operations
 Stronger Operational Visibility All 12 plants unified in a single real-time intelligence layer, eliminating report fragmentation.	 Improved Downtime Anticipation Predictive models surface failure risks days before they become costly incidents.	 Better Regional Benchmarking Plant-vs-plant comparison enables evidence-based capital allocation and maintenance prioritization.	 Organization-Wide Culture Shift Moving from a reactive reporting culture to a proactive, data-driven operational mindset.

REFLECTION

What I'd Do With Stakeholder Access

This project was designed using secondary research and assumption-based personas. With direct stakeholder access, I would ground every decision in validated user data.

- Validate KPI prioritization with actual VP stakeholders
- Test alert fatigue risk through usability testing sessions
- Conduct moderated testing on drill-down interaction patterns
- Refine AI recommendation logic with maintenance domain experts
- Measure decision efficiency improvements post-deployment

WHAT THIS DEMONSTRATES

Strategic UX Thinking
Designing for business outcomes, not just usability — connecting design decisions to organizational goals.

System-Level Architecture
Structuring complex information across four progressive layers that match real executive workflows.

Executive-Focused Hierarchy
Prioritizing 5-second scannability and financial context over granular operational detail.

Business-Aligned Visualization
Data visualizations tied directly to financial impact, not just operational reporting.

Enterprise Implementation
Every design decision validated against Power BI feasibility — bridging design vision with technical reality.